LEARNER GUIDE

Solving by elimination

Equations, Inequalities and Simultaneous Relationships · Lesson 8. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Two true equations can be **added** or **subtracted** to make another true equation — and if you choose well, one variable **cancels**. If a variable appears with **opposite signs** ($+y$ and $-y$), **adding** removes it; if it has the **same sign** (both $+y$), **subtracting** removes it. When the coefficients don't match, first **multiply** one equation through until they do. Once a variable is gone you have one equation in one unknown — solve it, find the other variable, and **check in both**. This is **elimination**.

What you are learning to notice

- Opposite signs on a variable → **add** to cancel it. (The easiest case to spot.)
- Same sign on a variable → **subtract** to cancel it.
- If coefficients don't match, **scale a row** (multiply the whole equation) until they do.
- After eliminating, solve, back-substitute, and check in *both* equations.

Moving through it, one step at a time

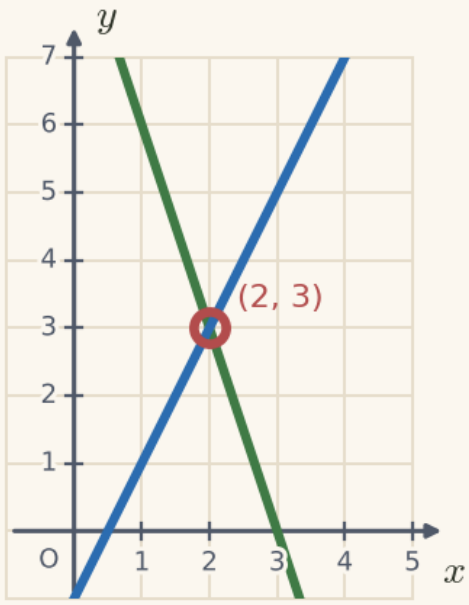
Line the two equations up. Look at each variable's signs and sizes. If one variable already matches, add (opposite signs) or subtract (same signs) to cancel it. If neither matches, multiply one equation through by a number so a variable's coefficients become equal, then add or subtract. Solve the resulting single equation, put the value back into an original equation to find the other variable, and check the pair in both.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Opposite signs — add

Solve $3x + y = 9$ and $2x - y = 1$. The y-terms are $+y$ and $-y$ — opposite.

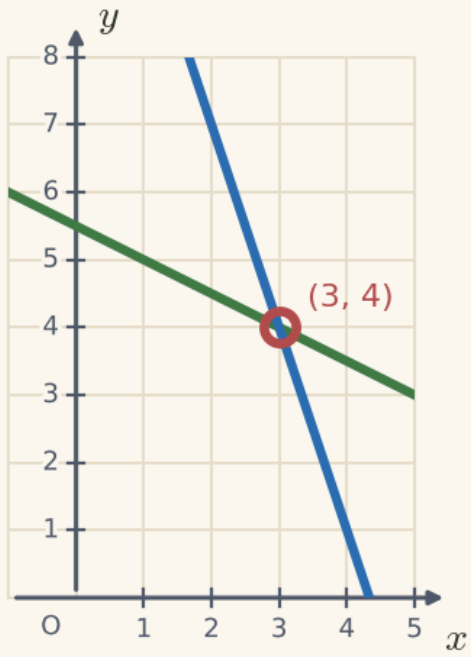


1. Add the two equations — the y cancels.	$(3x + y) + (2x - y) = 9 + 1$
2. One unknown remains.	$5x = 10$, so $x = 2$
3. Back-substitute to find y.	$3\times2 + y = 9$, so $y = 3$
4. Check in both.	$3\times2 + 3 = 9$ ✓ and $2\times2 - 3 = 1$ ✓

(2, 3) — opposite signs mean adding cancels y; then $x = 2$ and $y = 3$, true in both.

Example 2 — No match — scale, then subtract

Solve $x + 2y = 11$ and $3x + y = 13$. Nothing matches yet.



1. Multiply the second equation by 2 so the y-terms match.	$6x + 2y = 26$
2. Now both have $+2y$ — subtract to cancel y.	$(6x + 2y) - (x + 2y) = 26 - 11$
3. Solve the result.	$5x = 15$, so $x = 3$
4. Back-substitute and check.	$x + 2y = 11 \rightarrow y = 4$; both true ✓

(3, 4) — scaling the second equation by 2 lets subtraction cancel y; $x = 3$, $y = 4$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (adding and subtracting like terms, scaling an equation, and back-substituting to find the second variable), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.

- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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